Poisson regression

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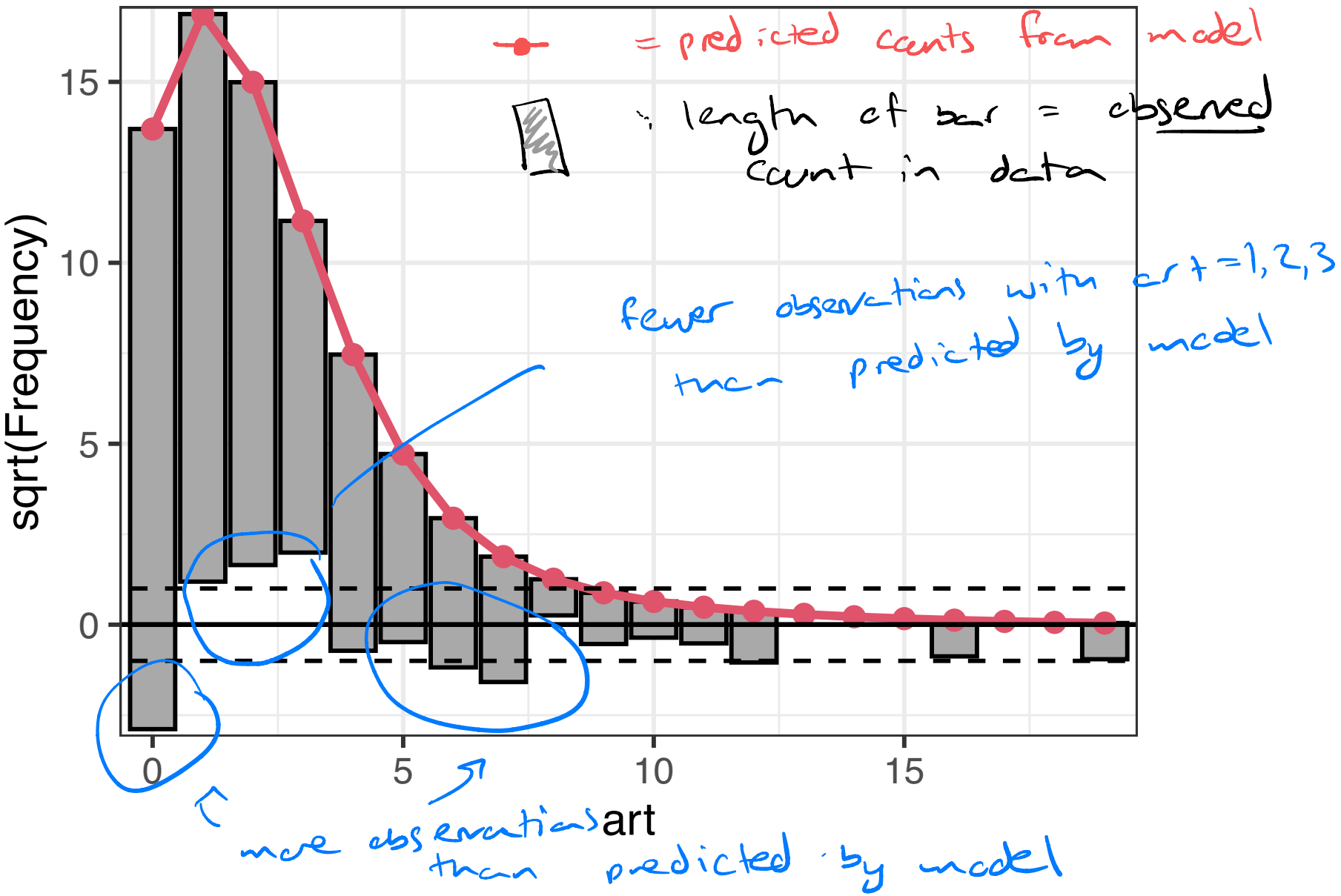
Last time: Activity

Use the provided packet to answer the following questions:

- Does the Poisson regression model appear to be a good fit?
- If not, which piece of the model do you think is mis-specified?

Rootograms: an overall assessment of model fit

```
topmodels::rootogram(m_art)
```



Rootograms: an overall assessment of model fit

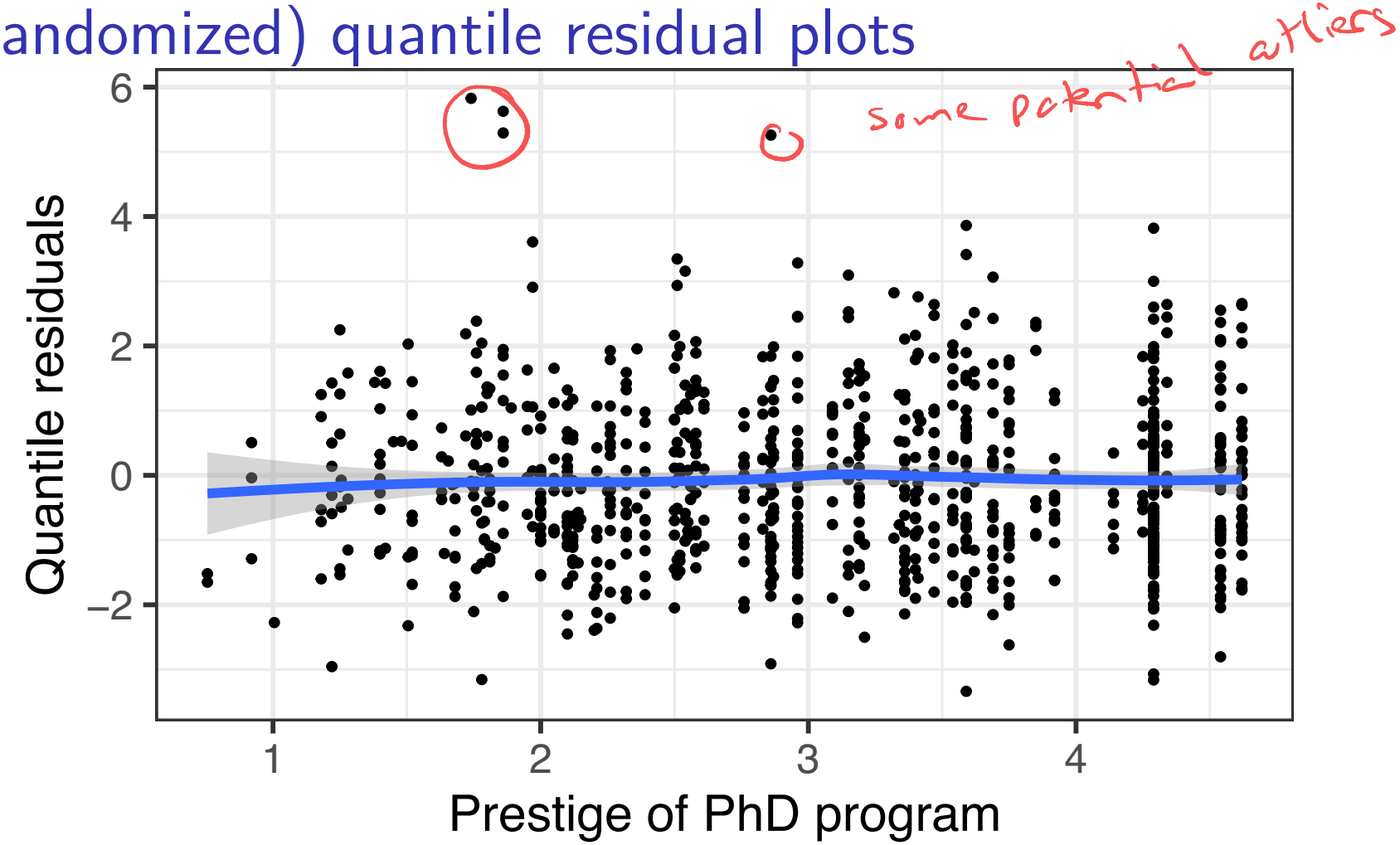
- horizontal axis: possible values for the response variable
- vertical axis (square root scale): predicted count for the model (points) and *observed* counts from the data (bars)
- Assessing whether predicted counts match observed counts
- Lack of fit could be due to:
  - Wrong choice of distribution
  - Additional explanatory variables need to be added to model
  - Issues with shape assumption

Question: We didn't make a rootogram for binary logistic regression. Why not?

$$u(\beta) = X^T(Y-\rho) = \sum_{i=1}^{n}(Y_i-\rho_i)\begin{bmatrix}1\\ x_{i1}\\ x_{i2}\\ \vdots\end{bmatrix}$$

$$\Rightarrow u(\hat{\beta}) = 0 \Rightarrow \sum_i (Y_i - \hat{\rho}_i) = 0$$

(Randomized) quantile residual plots



What I am looking for in a quantile residual plot?

- constant variance (non-constant variance ⇒ need different response distribution)
- random scatter around 0, no trend/pattern
- majority of residuals should be between −2 & 2 (since ~ N(0,1))

Examining influential points

Cook's distance: $C_i = \frac{1}{k+1} (\hat{\beta} - \hat{\beta}_{(i)})^T (\text{Var}(\hat{\beta}))^{-1} (\hat{\beta} - \hat{\beta}_{(i)})$

$(\beta \in \mathbb{R}^{k+1})$

$$\approx \frac{(y_i - \hat{\mu}_i)^2 h_i}{(k+1)\hat{\mu}_i (1-h_i)^2}$$

```
articles[which(cooks.distance(m_art) > 0.1),
         c("art", "kid5", "phd", "ment")]
```

```
##      art kid5  phd ment
## 328    1    1 1.78   77
## 914   16    0 1.74   21
## 915   19    0 1.86   42
```

probably not errors, but quite unusual!

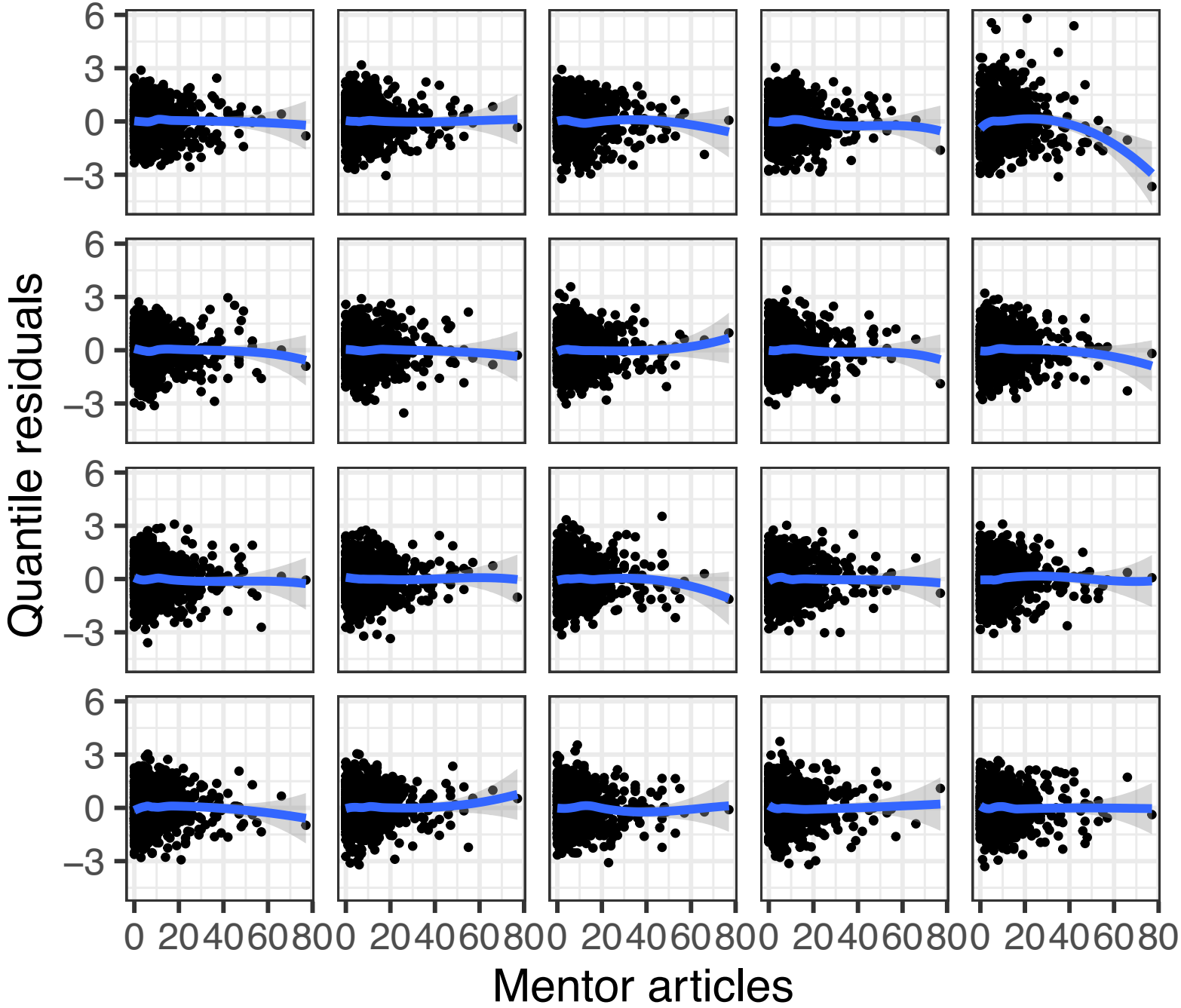
report results with and without unusual points

(Randomized) quantile residual plots



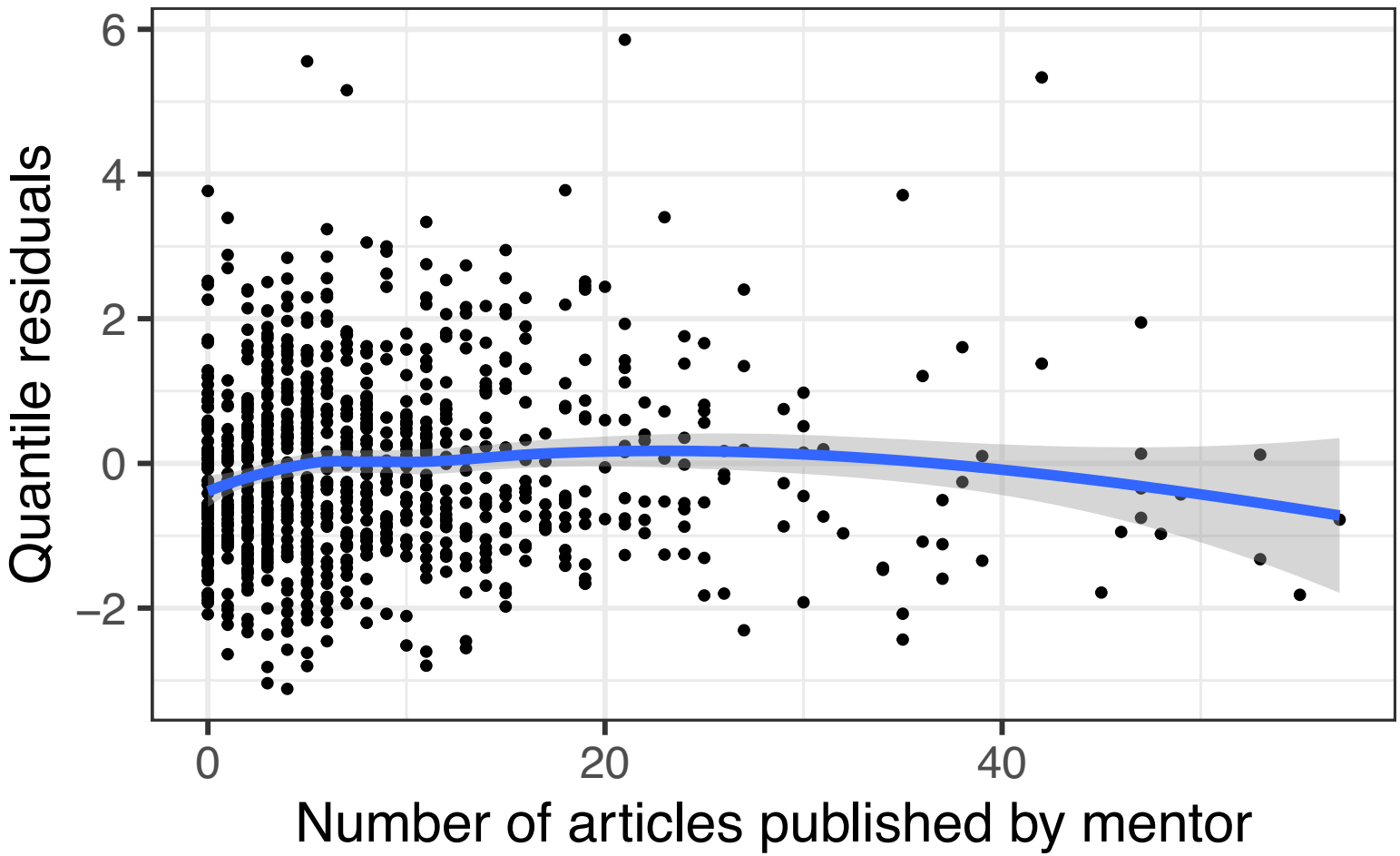
How could I decide whether deviations in the quantile residual plot are just due to random chance?

Plot lineup



Quantile residual plot

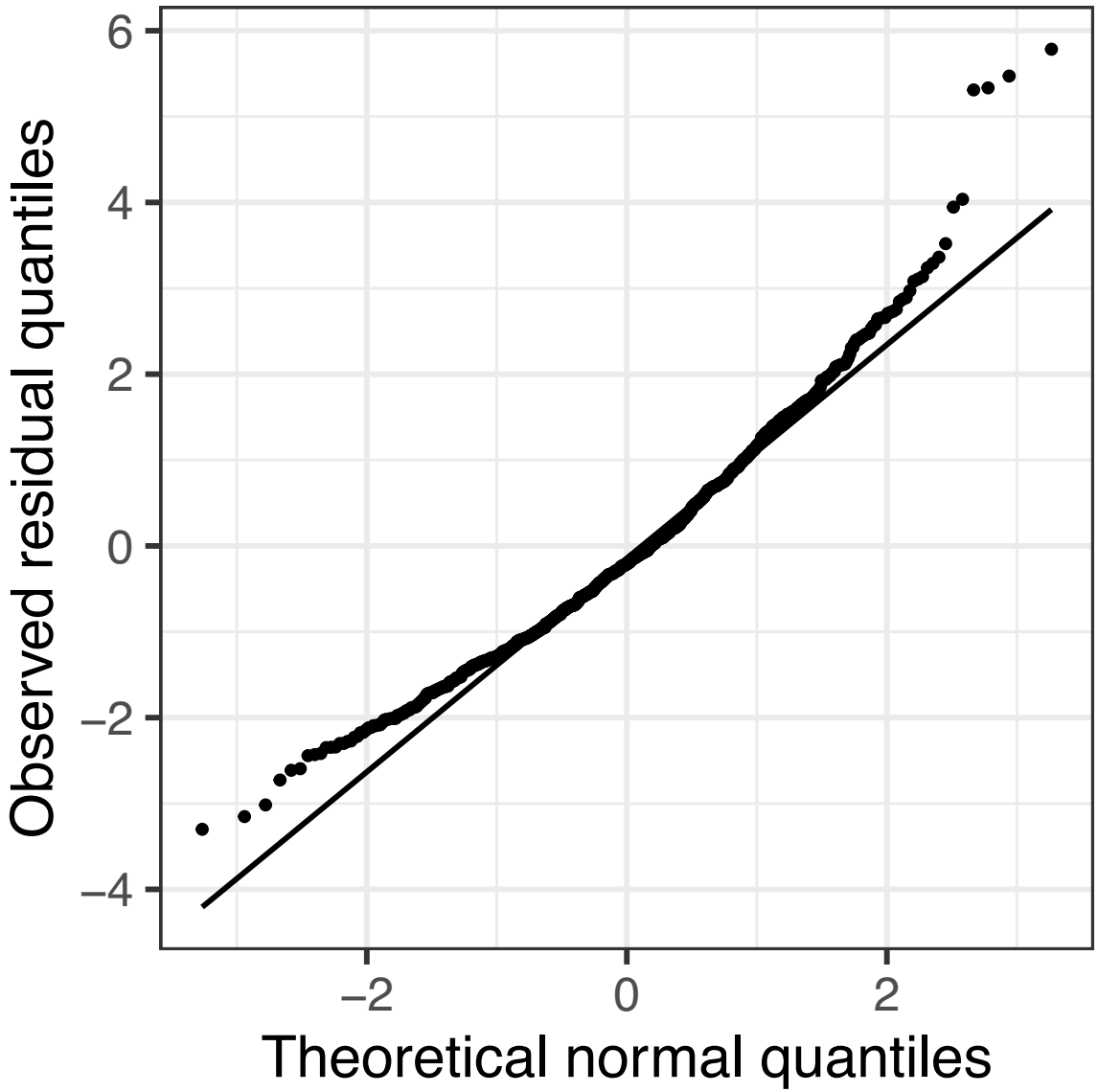
Plot without the two rows with the most number of mentor publications:



looks ok

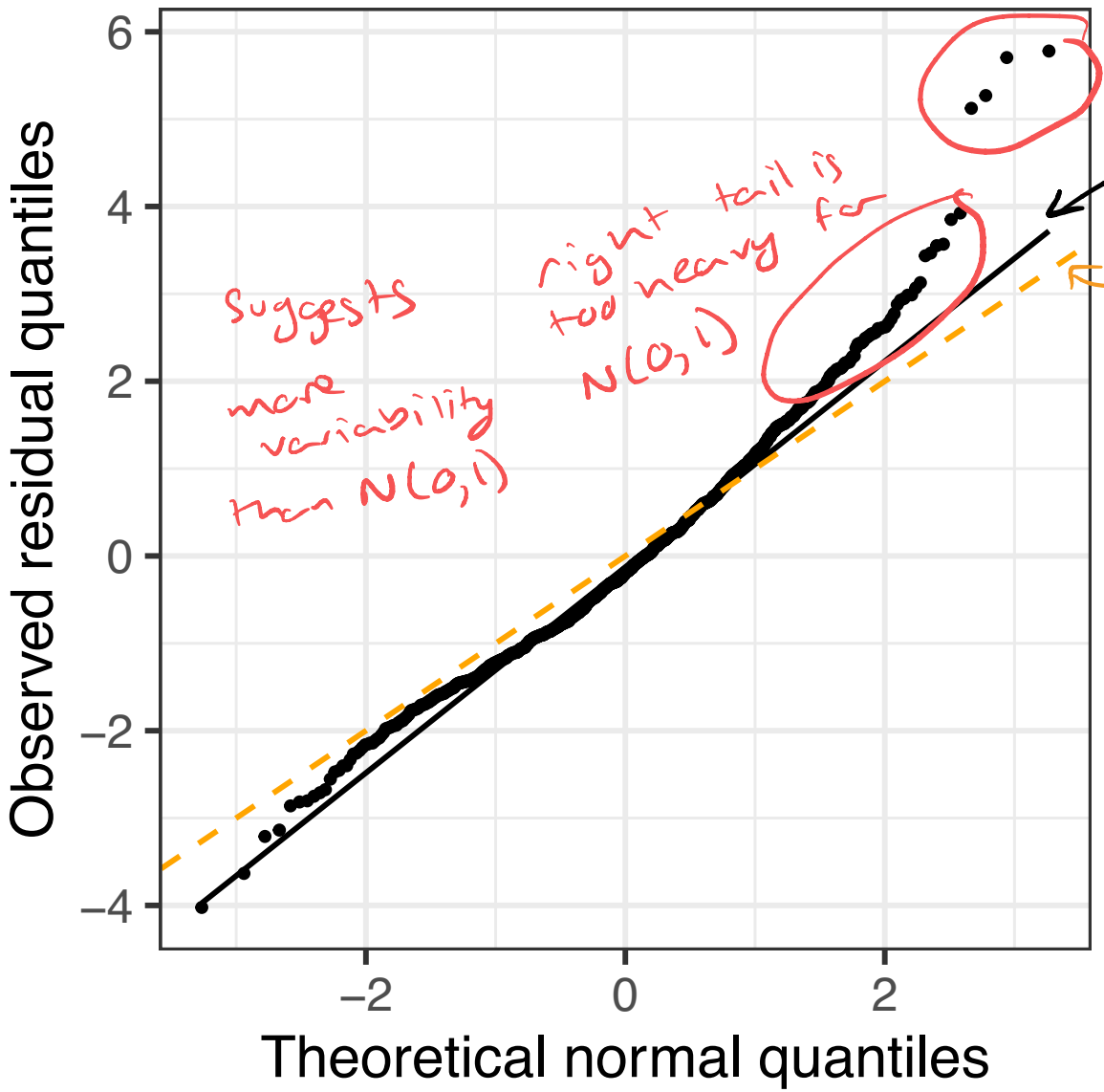
QQ plot

If the model is a good fit to the data, randomized quantile residuals should be approximately $N(0, 1)$.



QQ plot

If the model is a good fit to the data, randomized quantile residuals should be approximately $N(0, 1)$.



If the residuals are normal, should fall on some line (not necessarily $y = x$) depending on mean & variance

If residuals are $N(0,1)$, should fall on line $y = x$

Summary so far

For the Poisson regression on the articles data, here is my perspective:

- There are several potential outliers
- The rootogram shows our predicted counts are somewhat low in the tails of the distribution, compared to the observed counts
- Quantile residual plots suggest the shape assumption is overall pretty reasonable
- The QQ plot suggests that the quantile residuals are approximately normal, but variability of the residuals is higher than expected

Question: If this particular model is mis-specified, what do you think is the cause? How could we modify the model?